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# Supervised and Self-Supervised Learning on Food-251

FoodCNN under a 10M-parameter budget on a single 8GB GPU

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# The problem

# FoodX-251, under two hard constraints

- 251 fine-grained food categories, web-crawled **noisy** training labels, small clean validation set (Kaur et al., 2019)
- Fine-grained: inter-class differences (plating, garnish) can be **subtler** than intra-class variation (lighting, angle, portioning)
- Two constraints, not chosen for effect:
  - **From scratch** – no ImageNet or other pretraining
  - **Under 10,000,000 trainable parameters**, on a single **8GB** laptop GPU

# What's actually long-tailed here? Not the classes.

- Median 471 images/class, p10/p90 366/580 – close to **balanced**
- The famous 19.3x imbalance ratio is **one** 34-image outlier class
- **Class-balanced losses, logit adjustment, long-tail training: out of scope** – the real distributional problem is **label noise** (web-crawled train, clean val), not long tails

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**Protocol**

# Splitting the validation set, once, deterministically

- 11,994 val images -> 50/50 stratified split, fixed seed 251, every class within 1 image of exact half
- val-dev (6,063): every ablation, every architecture comparison
- val-test (5,931): touched **exactly once**, for the headline number
- Without this split, the headline number is selected-on
- 15-epoch fixed-seed proxy on val-dev (~18 min) stands in for a full 90-epoch run (~2h) during search; winner gets one full confirmation run

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# Architecture

# FoodCNN: spend the budget where it moves accuracy

## The trunk

- Stem -> 1/4 resolution, then 4 residual stages doubling width: **64-128-256-512**, blocks [2, 2, 2, 1]
- BN after **every** conv; zero-init each block's 2nd BN  $\gamma$ , so blocks start as identity – safe for a high LR
- ReLU **after** the residual add

## The one fix that mattered most

- (7, 7) map -> flatten -> Linear costs **6.4M params**: 94% of the model on its **least** expressive layer
- Global pool to (1, 1) first -> Linear(512, 251) = **129k**
- **6.58M params**, 66% of budget, 98% of it in the trunk

# Search by throughput and memory first, accuracy second

Screened trunk variants **synthetically** (no GPU-hours on accuracy yet):

- Stage widths must be **multiples of 32** – an off-alignment trunk has **fewer** params yet runs **46% slower** (falls off the tensor-core fast path)
- Final spatial resolution dominates cost more than any single knob – keeping a 6x6 map instead of 3x3 at **equal params** costs **2.9x** throughput and **2x** memory
- Lesson: hardware-aware screening before spending a single accuracy run

# Four candidates, one seed each

| Variant                   | Params | top-1  | top-5  | Peak VRAM |
|---------------------------|--------|--------|--------|-----------|
| [2,2,2,2] 64-448          | 9.46M  | 49.99% | 78.29% | 2.04 GiB  |
| [2,2,4,1] 64-512          | 8.94M  | 49.73% | 77.93% | 2.14 GiB  |
| baseline [2,2,2,1] 64-512 | 6.58M  | 48.82% | 77.12% | 1.99 GiB  |
| 5-stage 48-512            | 9.08M  | 47.42% | 75.94% | 1.61 GiB  |

- Top three within **1.2 points** – not enough to call on one seed
- Adopt **baseline**: statistically indistinguishable, **30% fewer params**, that headroom hadn't earned its keep yet
- A follow-up sweep **below** baseline found a dead heat down to 5.18M ( $p=1.0000$ ) and a cliff before 1.68M – capacity was never the binding constraint near baseline

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## Training recipe

# Six axes swept; only the learning rate paid

## Adopted recipe

- plain label-smoothed CE (0.1), SGD+Nesterov
- **lr 0.8**, batch 256, cosine + 5-epoch warmup
- bf16 autocast, channels\_last, 176px crop

## Swept on 15-epoch val-dev proxies

- **Learning rate** 0.05  $\rightarrow$  0.8: **+11.6 pts**, the one axis that paid
- **Augmentation:** TrivialAugment, RandAugment lose at 15 ep ( $p < 0.0001$ ); RandAugment **rematched at 30 ep** closes to 0.84 pts – **not** significant ( $p = 0.075$ )
- **Mixup, CutMix, EMA, GCE:** lose decisively ( $p < 0.0001$  each)
- **Batch size** 160-512: flat ( $p \geq 0.49$ )
- **Crop-scale floor:** 0.25 flat; 0.40 costs 1.17 pts ( $p = 0.023$ )

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# Results

# The full run

90 epochs, baseline trunk, tuned recipe, FixRes-style eval (train 176px, test 224px center crop):

| Split                                    | top-1         | top-3         | top-5         |
|------------------------------------------|---------------|---------------|---------------|
| val-dev, best                            | 64.36%        | 82.22%        | 87.51%        |
| <b>val-test (headline, touched once)</b> | <b>63.83%</b> | <b>81.79%</b> | <b>87.39%</b> |

- Train/val gap closes from **+11 points** at epoch 15 to **near zero** at epoch 90 – the model was underfitting, not overfitting; it needed the full schedule, not more regularization
- 2h05m wall clock, 1.99 GiB peak VRAM – comfortably under the 8GB ceiling

# Self-supervised track – in progress

- **Plan (decided):** SimSiam pretraining, 200 epochs @ 176px (~8 GPU-h), finetune (~1.8 GPU-h), against an **equal-GPU-hour supervised control** (~8 GPU-h) – without the control, “SSL helped” is indistinguishable from “more epochs helped, regardless of objective”
- SimSiam over BYOL: no momentum encoder, no second ~6.5M-param shadow network to fit in 8GB
- SimCLR excluded: degrades below batch sizes this budget can reach
- Scaffolding lands: `src/simsiam.py`, `--init-encoder`, tested on synthetic tensors – **the run itself has not happened yet**

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# Discussion

# Hardware measurement mattered as much as architecture

- Channel widths, spatial resolution, and **thermal throttling** moved results as much as any architectural choice
- GPU runs against a ~50W power cap; a short benchmark swung 45% between sweeps before fixing the timing methodology
- The full 90-epoch run itself ran **28% slower** than its own synthetic estimate – sustained load heat-soaks a laptop GPU in a way no short benchmark window reproduces
- Every throughput number here is an **upper bound**, and a **ratio** between variants – not a wall-clock promise

# Conclusion

- A plain residual CNN with a genuinely global pooling head reaches **63.83% top-1 / 87.39% top-5** on FoodX-251, from scratch, under 10M params, on a single 8GB GPU
- Two attractive-on-paper ideas – a fifth residual stage, a depthwise-separable trunk – both **lost once actually run**: 1.4-2.6 points and slower-and-heavier respectively
- Treating wall-clock and memory as design constraints, not just the parameter cap, is what got here
- **Open**: the self-supervised comparison against an equal-compute supervised control



Questions?